10.2 An example for a speci c w(t)

1. The book gives an example for a speci c piston motion:

$$w(t) = U(1 \quad e^{\quad t=\quad}) \qquad > 0 \tag{107}$$

where U is the nal speed of the piston and is a characteristic time of the piston acceleration.

2. integrating over time, one gets the position of the piston as:

$$X(t) = Ut + U\ (1 \quad e^{\quad t=\quad}) \tag{108}$$

which approaches the straight line $X(t) = U(t \quad)$ as $t >> \quad$, which is a valid boundary condition for the piston motion, that can be used to nd the function '(u) needed to nd the full solution for u(x; t) and c(x; t)!

$$X(t) = \quad c_0 \quad \frac{+1}{2} jw(t)j \quad t + '(jw(t)j) \tag{109}$$

substituting the assymptotic expression for X(t), one gets the velocity distribution with respect to x in the implicit form:

$$\boxed{x = \quad c_0 + \frac{+1}{2} u \quad t \quad u \ + \ (c \quad {}_0 + \frac{+1}{2} u + U) \ln \quad 1 + \frac{u}{U}} \tag{110}$$

which is valid according to the book in the region $X(t) < x < c\ {}_0 t$ and didn't understand why it isn't the smaller region $U(t \quad) < x < c \quad {}_0 t$.

2. Finding the Euler equations in terms of ξ:

$$\frac{D\rho}{Dt} = -\rho\frac{\partial u}{\partial x} \Rightarrow (u-\xi)\frac{d\rho}{d\xi} + \rho\frac{du}{d\xi} = 0 \tag{114}$$

$$\rho\frac{Du}{Dt} = -\frac{\partial p}{\partial x} \Rightarrow \rho(u-\xi)\frac{du}{d\xi} + \frac{dp}{d\xi} = 0 \tag{115}$$

$$\rho T\frac{DS}{Dt} = 0 \Rightarrow (u-\xi)\frac{dS}{d\xi} = 0 \tag{116}$$

3. and now the real magic happens. If we neglect the trivial solution of $u = \xi$ (hence constant fields and a uniform flow), then from the last equation we get that $S = const$ (isentropic flow) and therefore we can use the isentropic relation for dp to get:

$$dp = c^2 d\rho \Rightarrow \frac{dp}{d\xi} = c^2\frac{d\rho}{d\xi} \tag{117}$$

and then substituting the second equation into the first one to eliminate $\frac{d\rho}{d\xi}$, we get:

$$\left[(u-\xi)^2 - c^2\right]\frac{d\rho}{d\xi} = 0 \tag{118}$$

which means that $u - \xi = \pm c$ and so $\xi = u \pm c$. Substituting this relation inside the first equation, one gets:

$$\pm c\frac{\partial\rho}{\partial\xi} + \rho\frac{\partial u}{\partial\xi} = 0 \tag{119}$$

which can be rearranged to:

$$du \pm \frac{1}{\rho c}dp = 0 \tag{120}$$

which are exactly $dJ_\pm = 0$!!!

11.2 Analyzing the affect of the rarefaction wave over a single particle

1. The path of each particle in the rarefaction wave can be found by solving the ODE (with some initial condition x_0 at t_0):

$$\frac{dx}{dt} = u(x,t) \tag{121}$$

which essentially means integrating over the velocity field $u(x,t) = \frac{2}{\gamma-1}\left(c_0 - \frac{x}{t}\right)$. Since the right-moving wave propagates in the speed of c_0, one should notice that a particle initially at position x_0 will be affected by the wave only after time $t_0 = \frac{x_0}{c_0}$. This claim is illustrated in fig. 1.21 in the book.

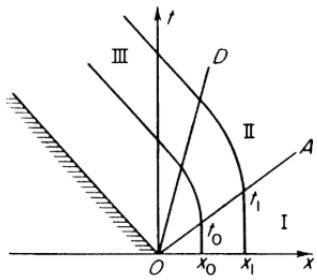


Fig. 1.21. Particle paths in x, t diagram for a centered rarefaction wave; OA is the wave head and OD is the wave tail.

2. It is also claimed that the faster the piston moves (larger $|U|$), the lower are c, p and ρ in the rarefaction wave (Eq. 1.53-1.56 in the book), which can be seen from the expressions for c, p and ρ (that all include negative expressions of $|U|$), and therefore there is a limit for the piston speed $|U| < \frac{2c_0}{\gamma-1}$ in order to keep c, p and ρ positive in the rarefaction wave.

3. However, there is a CAVEAT that says that physically, the piston speed can exceed that limit and thereby create a vacuum region in the gas. The particles will accelarate to the assymptotic speed of $|u|_{max} = \frac{2c_0}{\gamma-1}$ and the density and pressure will approach zero. This phenomenon is explained in pg. 42 in the book along with fig. 1.22.

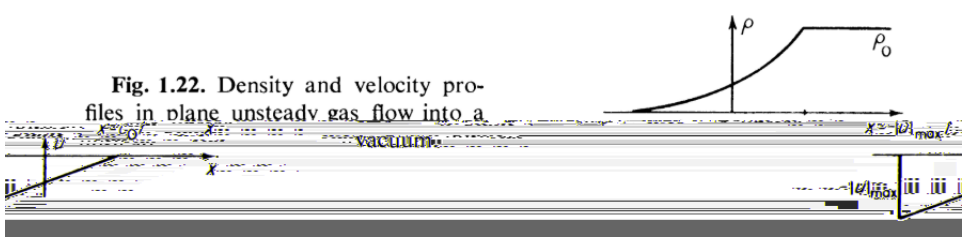


Fig. 1.22. Density and velocity profiles in plane unsteady gas flow into a vacuum.

I still need to make sure that I know how to derive these profiles from the equations.

11.3 Analyzing the vaccum solutio using Bernoullie's equation

1. The book discusses (pg. 42-43) the use of Bernoullie's equation in order to analyze the vacuum solution, where it uses the steady flow Bernoullie's equation along a streamline from the undisturbed gas to the point where the vacuum is formed. I didn't have time to write this part, it involves studying the Bernoullie's equation.

12 On the impossibility of the existence of a centered compression simple wave

1. The book discusses (pg. 43-44) the fact that a centered compression simple wave (the opposite of a rarefaction wave) is impossible, since it will require its tail to move faster than its head, which creates a graph that is opposite to fig 1.20 (the velocity profile of a centered rarefaction wave) and is actually non-physical.

2. I neglected here the full understanding of the calculation appearing in pg. 43 of the velocity profile. The figures are shown in the book.

13 Gas dynamics of shock waves

13.1 Problem setting and conservation equations

1. Mathematical background: In section 11 we showed that **since the problem is self-similar, the only solutions that satisfy the Euler equations are trivial uniform flows or centered simple waves**. However, in section 12 we showed that **a centered compression simple wave is *impossible***.

   Therefore, the only solution eliminates the unreal region II in fig 1.19 and creates a discontinuity (shock wave) between regions I and III.

2. initial gas parameters: The gas is initially at rest ($u = 0$) with uniform pressure p_0 and density ρ_0.

3. shock wave parameters: The piston moves at velocity u_1 (causing the particles trapped in the shock region to have the same velocity) and the shock wave moves to the right with speed D.

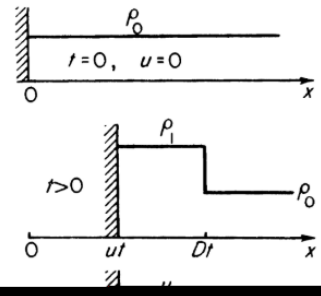


Fig. 1.25. Density and velocity profiles in a shock wave. The wave is produced by a piston which moves into the gas with a constant initial velocity. The top figure shows the initial state.

[redacted]

4. Conservation equations across the shock: The conservation equations across the shock (in the lab frame of reference) are:
   Mass conservation:
   At time t, the mass (per unit of one cross section) of the gas that already entered the shock region (looking at the top profile in Fig 1.25) is $\rho_0 Dt$. The mass of the gas that is actually inside the shock region (according to the lower figures in Fig 1.25) at time t is $\rho_1(D-u)t$.

$$\rho_0 Dt = \rho_1(D - u_1) \tag{122}$$

   Momentum conservation:
   THe total momentum of the gas at the shock region at time t is Mass $\times$ velocity $= \rho_0 Dt \cdot u$ (according to the Mass conservation equation). The change in momentum is caused by the net pressure (which is force per unit area of one cross section) acting over time t:

$$\rho_0 Dt = (p_1 - p_0)t \tag{123}$$

   Note that the pressure p_1 is the pressure that the gas in the shock region exerts on the environment (to the right).

   Energy conservation:
   Here the argument says that the change in the total energy of the gas in the shock region at time t is Mass $\times$ (change in internal energy + kinetic energy) $= \rho_0 Dt\left(\varepsilon_1 + \frac{u^2}{2}\right)$. This should be equal to the work done by the gas at pressure region over time t which is $p_1 ut$.

$$\rho_0 Dt\left(\varepsilon_1 - \varepsilon_0 + \frac{u^2}{2}\right) = p_1 ut \tag{124}$$

13.2 Changing to the shock frame of reference

1. It is often more convenient to analyze the shock wave in the shock frame of reference, where the shock is stationary and the **gas flows into the shock at speed** $u_0 = -D$ from the right and **exits the shock at speed** $u_1 = -(D - u)$ to the left.

2. Thus, the conservation equations in the shock frame of reference are:

$$\text{Mass conservation:} \quad \rho_0 u_0 = \rho_1 u_1 \tag{125}$$

$$\text{Momentum conservation:} \quad p_0 + \rho_0 u_0^2 = p_1 + \rho_1 u_1^2 \tag{126}$$

$$\text{Energy conservation:} \quad \varepsilon_0 + \frac{p_0}{\rho_0} + \frac{u_0^2}{2} = \varepsilon_1 + \frac{p_1}{\rho_1} + \frac{u_1^2}{2} \tag{127}$$

   there is also a reference to the Bernoullie's equation for steady flow along a streamline, which is essentially the same as the energy conservation equation above:

$$h + \frac{u^2}{2} = \varepsilon + \frac{u^2}{2} + \frac{p}{\rho} = const \tag{128}$$

3. <u>Derivation of the conservation equations</u>:
   These equations can actually be achieved through integrating over the Euler equations in their "Flux" form (the flux for mass, momentum and energy) across the shock discontinuity. This is shown through Eqs. (1.65) in the book:

$$\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial x}(\rho u) \tag{129}$$

$$\frac{\partial (\rho u)}{\partial t} = -\frac{\partial}{\partial x}(p + \rho u^2) \tag{130}$$

$$\frac{\partial}{\partial t}\left(\rho\varepsilon + \frac{1}{2}\rho u^2\right) = -\frac{\partial}{\partial x}\left[\rho u\left(\varepsilon + \frac{1}{2}u^2 + \frac{p}{\rho}\right)\right] \tag{131}$$

14 Huguniot Curves

14.1 The derivation of the Huguniot Curve

1. First, one should rearrange the mass equation through the definition of the specific volume $V = \frac{1}{\rho}$ to get:

$$\frac{V_0}{V_1} = \frac{u_0}{u_1} \tag{132}$$

Then, eliminating u_0 and u_1 from the momentum equation using the mass equation (with the specific volume), one gets:

$$u_0^2 = V_0^2 \frac{p_1 - p_0}{V_0 - V_1} \tag{133}$$

$$u_1^2 = V_1^2 \frac{p_1 - p_0}{V_0 - V_1} \tag{134}$$

Substituting this relations into the energy equation, one gets the Huguniot equation:

$$\varepsilon_1 - \varepsilon_0 = \frac{1}{2}(p_1 + p_0)(V_0 - V_1) \tag{135}$$

which is often denoted using the enthalpy $h = \varepsilon + pV$ as:

$$h_1 - h_0 = \frac{1}{2}(p_1 - p_0)(V_0 + V_1) \tag{136}$$

which in general can be written as:

$$p_1 = H(V_1; p_0, V_0) \tag{137}$$

2. The Huguniot equation describes the possible end states (p_1, V_1) that can be achieved through a shock wave from a given initial state (p_0, V_0).

3. In that manner, it is analogous to the isentropic relations that describe the possible end states that can be achieved through isentropic processes from a given initial state throgh the isentropic relation $p_1 = P(V_0, S)$. However, the set of possible isentropic curves is parametered using 1D parameter S (entropy), while the Huguniot curves are parametered using 2D parameters (p_0, V_0).

15 Shock waves in an ideal gas with constant specific heats

15.1 Analyzing the Huguniot curves

1. The internal energy and enthalpy for an ideal gas with constant specific heats are:

$$\varepsilon = c_v T = \frac{pV}{\gamma - 1} \tag{138}$$

$$h = c_p T = \frac{\gamma pV}{\gamma - 1} \tag{139}$$

and therefore they can be directly substituted into the Huguniot equation (Eq. 136) to get:

$$\frac{\gamma}{\gamma-1}(p_1V_1 - p_0V_0) = \frac{1}{2}(p_1 - p_0)(V_0 + V_1) \tag{140}$$

which can be rearranged to:

$$\frac{p_1}{p_0} = \frac{(\gamma+1)V_0 - (\gamma-1)V_1}{(\gamma+1)V_1 - (\gamma-1)V_0} \tag{141}$$

2. For some reason (that the book says will become clear at section 1.17), the book calims that the Huguniot curve is physically unattainable for $V_1 > V_0$ (expansion shock) and shows it as a dashed line in fig. 1.27

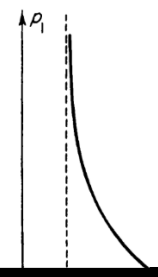


Fig. 1.27. Hugoniot curve.

[redacted]

3. The book also considers the ratio of densities or specific heats (they are inversely proportional) that are easily produced from the Hugoniot equation (Eq. 141):

$$\frac{\rho_1}{\rho_0} = \frac{V_0}{V_1} = \frac{(\gamma+1)p_1 + (\gamma-1)p_0}{(\gamma-1)p_1 + (\gamma+1)p_0} \tag{142}$$

Even if $p_1 \gg p_0$, the density ratio is bounded by:

$$\frac{\rho_1}{\rho_0} < \frac{\gamma+1}{\gamma-1} \tag{143}$$

and the book discusses that in monoatomic ideal gas this ratio equals 4, in diatomic ideal gas it equals 6 and even at gases with high temperatures, vibrational modes, ionization and variating specific heats, this ratio is still bounded and cannot exceed 11-13.

15.2 Limits of strong & weak shocks

1. As we have seen, the density ratio across the shock is bounded. When $p_1 \gg p_0$ and $\frac{V_1}{V_0} \approx \frac{\gamma-1}{\gamma+1}$, the increase in temperature is proportional to the pressure increase:

$$\frac{T_1}{T_0} = \frac{p_1 V_1}{p_0 V_0} \approx \frac{(\gamma-1)}{(\gamma+1)} \frac{p_1}{p_0} \tag{144}$$

2. The equations for the ratio $\frac{V_0}{V_1}$ and the equations for the velocities 134 can be rearranged to get:

$$u_0^2 = \frac{V_0}{2} \left[(\gamma-1)p_0 + (\gamma+1)p_1\right] \tag{145}$$

$$u_1^2 = \frac{V_0}{2} \frac{\left[(\gamma+1)p_0 + (\gamma-1)p_1\right]^2}{\left[(\gamma-1)p_0 + (\gamma+1)p_1\right]^2} \tag{146}$$

which in the limit of strong shocks ($p_1 \gg p_0$) become:

$$u_0^2 \approx \frac{(\gamma+1)}{2} p_1 V_0 \tag{147}$$

$$u_1^2 \approx \frac{(\gamma-1)^2}{2(\gamma+1)} p_1 V_0 \tag{148}$$

knowing that the speed of sound in the gas is:

$$c_{0,1}^2 = \gamma p_{0,1} V_{0,1} \tag{149}$$

one can write the Mach numbers of the flow into and out of the shock as:

$$M_0^2 = \frac{u_0^2}{c_0^2} \approx \frac{(\gamma+1)}{2\gamma} \frac{p_1}{p_0} \tag{150}$$

$$M_1^2 = \frac{u_1^2}{c_1^2} \approx \frac{(\gamma-1)}{2\gamma} \tag{151}$$

which are exactly Eq. (1.83) and (1.84) in the book under the limit $p_1 \gg p_0$.

3. In a discussion at the top of pg. 53 in the book, it claims that in the limit of weak shocks ($p_1 \approx p_0$), Eq. (1.83) and (1.84) still hold (which I don't understand, because they assumed Eq. (1.82) that assumes $p_1 \gg p_0$) the Mach number become close to unity (i.e. $u_0 \approx c_0 \approx c_1 \approx u_1$) and the shock wave becomes a weak discontinuity that can be analyzed using the linear acoustic theory (since the perturbations in pressure, density and velocity are small).

4. Physically speaking, the gas enters the shock region at supersonic speed ($M_0 > 1$) and exits the shock region at subsonic speed ($M_1 < 1$). The Mach number of the supersonic flow into the shock can be arbitrarily large (strong shock) while the Mach number of the subsonic flow ($M_0 \to \infty$) out of the shock is bounded by $M_1 \geq \sqrt{\frac{\gamma-1}{2\gamma}}$ which is less than 1 for every $\gamma > 1$.

15.3 Increase in Entropy

1. The book discusses (pg. 53 bottom - 54) the fact that the entropy increases across the shock wave according to the formula:

$$\begin{aligned} S_1 - S_0 &= c_v \ln \left[\frac{p_1}{p_0} \left(\frac{V_1}{V_0} \right)^\gamma \right] \\ &= c_v \ln \left[\frac{(\gamma+1)p_1 + (\gamma-1)p_0}{(\gamma-1)p_1 + (\gamma+1)p_0} \left(\frac{(\gamma+1)V_0 - (\gamma-1)V_1}{(\gamma+1)V_1 - (\gamma-1)V_0} \right)^\gamma \right] \end{aligned} \tag{152}$$

   from which one can see that for weak shocks ($p_1 \approx p_0$), the entropy increase is very small (second order in the pressure difference), and for strong shocks ($p_1 \gg p_0$), the entropy increase is large and proportional to $\ln(p_1/p_0)$, which is unbounded.

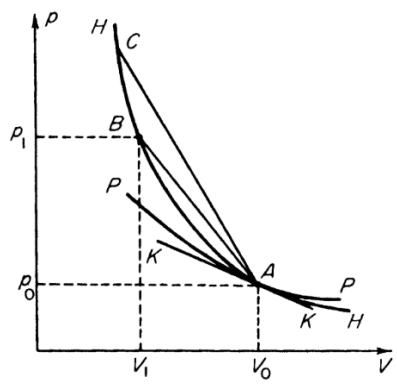
2. The book says that the increase in entropy happens due to the irreversible nature of the shock wave, and is derived from the conservation laws. It says that **the rate of the increase in entropy** is not determined by the Euler equations, but rather by the **dissipative processes** (like viscosity and heat conduction) that rely on length scales which are the mulecular mean free path that is neglected in the discussion of the Euler equations.

16 Geometric interpretation of the shock relations

16.1 The relation between the Hugoniot and Isentrope curves

1. Suppose a gas initially at state $A(p_0, V_0)$. This state defines the curve of possible end states that can be achieved through isentropic processes (the isentrope curve) and the curve of possible end states that can be achieved through shock waves (the Hugoniot curve), shown in Fig. 1.28.

Fig. 1.28. A p, V diagram. HH is a Hugoniot curve, PP is an isentrope, and KK is a tangent to both the Hugoniot and isentrope at the initial state represented by the point $A(V_0, p_0)$.



2. Both the Hugoniot and Isentrope curves are convex downwards, and they are tangent at point A (proper calculation ahead).

3. One notices that another point B(p_1, V_1) symbolizes another state that can be achieved through a shock wave from state A, and the point P(p_1, V_2) symbolizes the state that can be achieved through an isentropic process from state A.

4. From the velocity equation 134 (Eq. 1.67 in the book), one can write the shock speed squared as the negative of slope of the secant line AB:

$$D^2 = u_0^2 = -V_0^2 \frac{(p_1 - p_0)}{V_1 - V_0} \tag{153}$$

this formula shows explicitly that the higher final pressure $p_1 \rightarrow$ the higher shock speed D.

5. Using Eq. (1.75) in the book (which is the ratio of specific volumes across the shock), one can calculate the derivative of the curve:

$$\frac{dp_1}{dV_1} = \frac{d}{dV_1}\left[\frac{(\gamma+1)V_0 - (\gamma-1)V_1}{(\gamma+1)V_1 - (\gamma-1)V_0} p_0\right] \tag{154}$$

$$= -\frac{(\gamma-1)p_0}{(\gamma+1)V_1 - (\gamma-1)V_0} - \frac{(\gamma+1)V_0 - (\gamma-1)V_1}{[(\gamma+1)V_1 - (\gamma-1)V_0]^2}(\gamma+1)p_0 \tag{155}$$

By setting $V_1 = V_0$ in the above equation, one gets the slope of the Hugoniot curve at point A:

$$\left(\frac{dp}{dV}\right)_{Hugoniot,A} = -\frac{\gamma p_0}{V_0} \tag{156}$$

which is exactly the slope of the isentrope curve at point A ($PV^\gamma \sim const \Rightarrow \left(\frac{\partial p}{\partial V}\right)_S = -\frac{\gamma p}{V} wg7$)

6. One can notice that the shock velocity under the limit of weak shocks ($p_1 \approx p_0$) is:

$$D^2 = -V_0^2 \frac{p_1 - p_0}{V_1 - V_0} \rightarrow \left(\frac{dp}{dV}\right)_{Hugoniot,A} = \gamma p_0 V_0 = c_0^2 \tag{157}$$

and also $D = u_0 > c_0$ (supersonic flow into the shock) as expected since the curve is convex downwards. It means that the initial slope of the Hugoniot curve at point A is defined by the speed of sound in the gas.

7. In section 1.18 we will show that also the second derivative of the Hugoniot and Isentrope curves are equal at point A, which means that the two curves are **tangent up to second order at point A**.

16.2 The difference between the shock and isentropic processes as a change in entropy

1. One should notice that the change in the specific energy during the shock compression form state A to state B is:

$$\Delta\varepsilon_{AB} = \varepsilon_1 - \varepsilon_0 = \frac{1}{2}(p_1 + p_0)(V_0 - V_1) \tag{158}$$

according to Eq. 135 (Eq. 1.71 in the book). This is exactly the area of the trapezoid MABN in fig. 1.29 in the book.

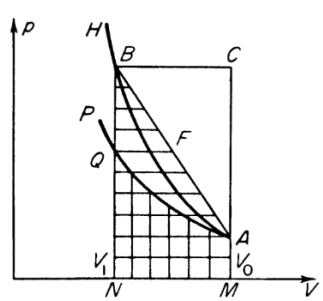


Fig. 1.29. Geometrical interpretation of the energy increase across a shock wave. *H* is the Hugoniot curve and *P* is the corresponding isentrope.

2. One can imagine an isentropic process from state A to state P, followed by a heating process from state P to state B (vertical line, isochoric processes). The total energy change in this process is:

$$\Delta\varepsilon_{APB} = \int_{S_0}^{S_1} TdS = \varepsilon_1 - \varepsilon' \tag{159}$$

where we sticked here to the book's notation where ε' is the specific energy at point P.

3. Another cool property is the area of the rectangle MNBC, which is equal to $p_1(V_0 - V_1)$ which is exactly the total work done on a unit mass of gas during the piston compression from state A to state B. Across a strong shock when $p_1 \gg p_0$, the area of the rectangle is approximately twice the area of the trapezoid, which means that half of the work done on the gas is converted to an increase in internal energy, while the other half is dissipated as an increase in entropy. One can see that from the change in kinetic energy per unit mass across the shock:

$$\frac{u^2}{2} = \frac{(u_0 - u_1)^2}{2} = \frac{1}{2}\left(\frac{p_1 - p_0}{V_0 - V_1}\right) \to \frac{1}{2}p_1(V_1 - V_0) \quad \text{for } p_1 \gg p_0 \tag{160}$$

16.3 Two successive shocks

1. Suppose that the gas is initially at state $\mathrm{A}(p_0, V_0)$, then it is compressed through a shock wave to state $\mathrm{B}(p_1, V_1)$ on H_A curve, and then it is compressed again through another shock wave to another point in the illustrated H_B. This is shown in fig. 1.30 in the book.

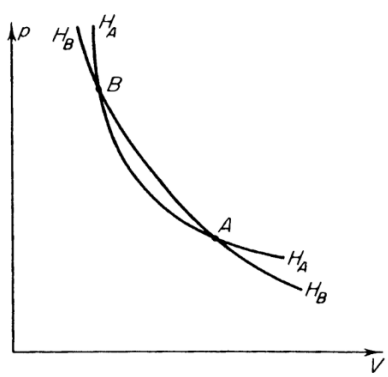


Fig. 1.30. A p, V diagram, illustrating the relationship between the gas velocity and the speed of sound in a shock wave.

Since the Hugoniot function $p_1 = H(V_1; p_0, V_0)$ is symmetric to the permutation $1 \leftrightarrow 0$ (can be seen from Eq. 136), the point A also lies on the Hugoniot curve H_B.

2. Another interesting point that arises from this graph is the fact that $u_1 < c_1$ (subsonic flow out of the shock). It is seen from these equations:

$$u_1^2 = -V_1^2 \frac{p_1 - p_0}{V_1 - V_0} \tag{161}$$

$$c_1^2 = -V_1^2 \left(\frac{\partial p}{\partial V}\right)_{H_B, S} \tag{162}$$

and it is seen from the graph that the slope of the tangent line at point B is steeper than the slope of the secant line AB, which means that $u_1^2 < c_1^2$ or $M_1 < 1$ as expected.

3. The book also comments about the physical reason for the feature that H_B passes to the left of H_A at pressures higher than p_1. It says that since the volume in a shock compression is bounded by $V_B > \frac{\gamma-1}{\gamma+1} V_A$, the second shock can decrease the volume to a state C with a lower boundary which is $V_C > \frac{\gamma-1}{\gamma+1} V_B > \left(\frac{\gamma-1}{\gamma+1}\right)^2 V_A$.

4. At the top of pg. 59, the book comments that it's impossible to reach a final state (p_1, V_1) through two successive shocks that is different from the state achieved through a single shock from the initial state (p_0, V_0). This process is shown in fig. 1.31 in the book.

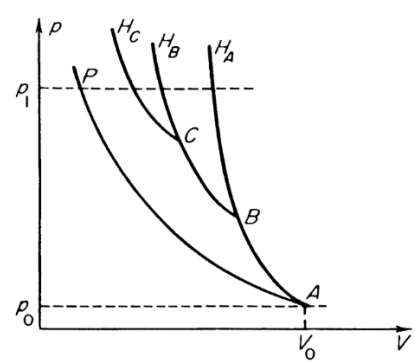


Fig. 1.31. Single and multiple shock and isentropic compression of a gas to the same final pressure p_1. H_A, H_B, and H_C are Hugoniot curves with the initial states A, B, and C, respectively. P is an isentrope.

The book also says that this fact can be proven using the fact that the Hugoniot curves are function of two parameters (p_0, V_0) and I need to understand why (optional though).

17 On the impossibility of a rarefaction shock wave in a normal fluid

17.1 Analysis using the second law of thermodynamics

1. Inequalities (1.86) & (1.87) in the book show the difference in the discontinuity relations between shock waves and rarefaction shocks. For shock waves - $p_1, \rho_1, S_1 > p_0, \rho_0, S_0$ while for rarefaction shocks - $p_1, \rho_1, S_1 < p_0, \rho_0, S_0$. This is shown by Fig. 1.32 in the book.

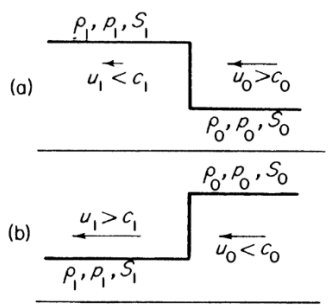


Fig. 1.32. Schematic representation of (a) compression and (b) rarefaction shock waves. The gas flows into the discontinuity from right to left.

The book emphasizes the fact that the conservation laws themselves **do not forbid the existence of rarefaction shocks**, but rather the second law of thermodynamics (the increase in entropy) forbids their existence in normal fluids - the explanation is given through the analysis of fig. 1.33 where the final state B lies on the right side (bigger volume) of the initial state A on the Hugoniot curve, which means that $S_1 < S_0$, e.g. a decrease in entropy.

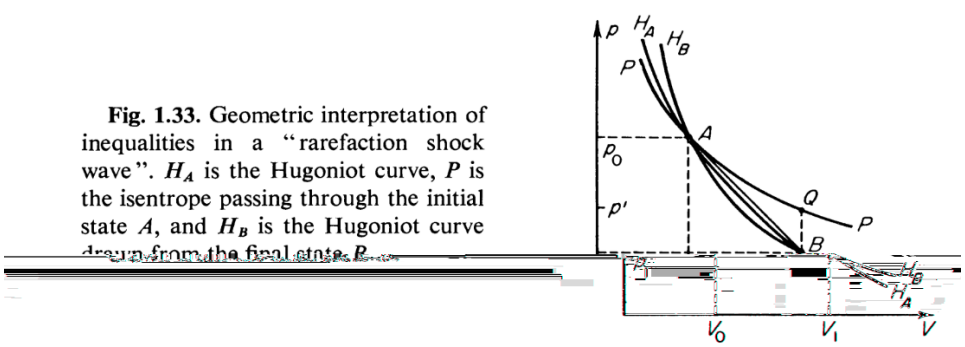


Fig. 1.33. Geometric interpretation of inequalities in a "rarefaction shock wave". H_A is the Hugoniot curve, P is the isentrope passing through the initial state A, and H_B is the Hugoniot curve drawn from the final state B.

2. The book says that this phenomenon is general and applies to all fluids where the Hugoniot curves are convex downwards in the $p - V$ plane, e.g. $\left(\frac{\partial^2 p}{\partial V^2}\right)_H > 0$ (these are called "normal fluids"), and it says that this statement will be proven for weak shockwaves in section 1.18.

17.2 Explanation using Mechanical Stability

1. The book discusses (pg. 61-62) the mechanical stability argument that explains the impossibility of rarefaction shocks in normal fluids.

2. It says (mid pg. 62) that a wave that propagates through a fluid supersonically is "stable" in the sense that small perturbations ahead of the wave cannot affect the wave structure, while a wave that propagates subsonically is "influencable" in the sense that small perturbations ahead of the wave can affect the wave structure.

3. Therefore, a shock wave is stable regarding small perturbations that occur behind the shock front (because it propagates through the undisturbed gas in a supersonic speed), while it's strength **is** influencable by small perturbations that occur behind it since it propagates through the disturbed gas in a subsonic speed (which explains the fact that heating/cooling or compressing/expanding the gas behind the shock can affect its strength). This means that strength is influencable in a shock wave, but the wave structure is stable.

4. On the other hand, the strength of a rarefaction shock is stable (since it propagates through the undisturbed gas in a supersonic speed), but its structure is influencable (since it propagates through the disturbed gas in a subsonic speed). This means that the rarefaction shock structure is not stable as a discontinuity, and it's also not physical.

17.3 On the relation between entropy to stability

The book actually showed the significant result that can be summarized using the following relation by ChatGPT:

$$\text{Entropy Increase} \iff \text{Causal Control of the Shock} \iff \text{Wave Structure Stability} \tag{163}$$

This conclusion is also spectacularly summarized in table 1.1 in the book, that relates it to the section of the book where it is discussed rarefaction smooth waves (section 1.12) and centered compression simple waves (section 1.13).

Table 1.1

ATTAINABLE STATES BY MEANS OF COMPRESSION AND RAREFACTION WAVES FOR NORMAL FLUIDS

	Compression wave	Rarefaction wave
Discontinuity	Possible; entropy increases; mechanically stable	Impossible; entropy decreases; mechanically unstable
Smooth distribution	Impossible[a]; unlimited increase in the steepness of the wave front with the final result of "overshooting"	Possible; the distribution becomes increasingly smoother with time

18 Weak shock waves

18.1 Derivation of the impossibility of a rarefaction shock wave for a weak shock in a general normal gas

1. The book asssumes now a general gas (with a general function $\varepsilon(S, V)$) and analyzes weak shocks where $\varepsilon_1 - \varepsilon_0$ is small so the Hugoniot curve can be expanded in Taylor series around point A(p_0, V_0):

2. The theoretical equation for a Hugoniot curve is:

$$\varepsilon_1 - \varepsilon_0 = \frac{1}{2}(p_1 + p_0)(V_0 - V_1) \tag{164}$$

The Taylor expansion of the LHS around point A (for 1st order in ΔS and 3rd order in ΔV) is:

$$\varepsilon_1 - \varepsilon_0 = \left(\frac{\partial \varepsilon}{\partial S}\right)_V (S_1 - S_0) + \left(\frac{\partial \varepsilon}{\partial V}\right)_S (V_1 - V_0) + \frac{1}{2}\left(\frac{\partial^2 \varepsilon}{\partial V^2}\right)_S (V_1 - V_0)^2 + \frac{1}{6}\left(\frac{\partial^3 \varepsilon}{\partial V^3}\right)_S (V_1 - V_0)^3 + ... \tag{165}$$

which according to the thermodynamic relation $d\varepsilon = TdS - pdV$ can be written as:

$$\varepsilon_1 - \varepsilon_0 = T_0(S_1 - S_0) - p_0(V_1 - V_0) - \frac{1}{2}\left(\frac{\partial p}{\partial V}\right)_S (V_1 - V_0)^2 - \frac{1}{6}\left(\frac{\partial^2 p}{\partial V^2}\right)_S (V_1 - V_0)^3 + ... \tag{166}$$

The RHS can be expanded as (to the same orders as):

$$\frac{1}{2}(p_1+p_0)(V_0-V_1) = \left(\frac{\partial p}{\partial S}\right)_V (S_1-S_0)(V_0-V_1) - p_0(V_1-V_0) - \frac{1}{2}\left(\frac{\partial p}{\partial V}\right)_S (V_1-V_0)^2 - \frac{1}{4}\left(\frac{\partial^2 p}{\partial V^2}\right)_S (V_1 \tag{167}$$

If we neglect the term $\left(\frac{\partial p}{\partial S}\right)_V (S_1 - S_0)(V_0 - V_1)$ under the argument that $\Delta S \sim \Delta V^3$ (which will be consistent with the final result), we can equate the two expansions to get:

$$T_0(S_1 - S_0) = \frac{1}{12}\left(\frac{\partial^2 p}{\partial V^2}\right)_S (V_1 - V_0)^3 \tag{168}$$

which finally shows that with normal gases (with $\left(\frac{\partial^2 p}{\partial V^2}\right)_S > 0$) the entropy increases across a weak shock wave ($S_1 > S_0$), while rarefaction shocks ($V_1 > V_0$) are impossible since they would cause a decrease in entropy ($S_1 < S_0$). More explanation about it (Including the neuances about what the definition of a "normal" gas actually means and how it relates to the coefficient of thermal expansion at constant pressure) is given at the bottom of pg. 64.

18.2 Relation between the expansion of the Hugoniot and Isentrope curves at point A

1. One can expand the pressure along the isentrope & Hugoniot curves around point A in Taylor series up to third order in ΔV:

$$p_{Isentrope} = p_0 + \left(\frac{\partial p}{\partial V}\right)_S (V_1 - V_0) + \frac{1}{2}\left(\frac{\partial^2 p}{\partial V^2}\right)_S (V_1 - V_0)^2 + \frac{1}{6}\left(\frac{\partial^3 p}{\partial V^3}\right)_S (V_1 - V_0)^3 + ... \tag{169}$$

$$p_{Hugoniot} = p_0 + \left(\frac{\partial p}{\partial V}\right)_S (V_1 - V_0) + \frac{1}{2}\left(\frac{\partial^2 p}{\partial V^2}\right)_S (V_1 - V_0)^2 + \frac{1}{6}\left(\frac{\partial^3 p}{\partial V^3}\right)_S (V_1 - V_0)^3 + ... \tag{170}$$

   where the third-order term in the Hugoniot expansion includes the linear term in ΔS as should be considered from the previous section. Need to FIX!!!! And understand BTW

2. That being said, one can see that the first and second derivatives of the Hugoniot and Isentrope curves are equal at point A, while the third derivatives are different by the term:

$$\left(\frac{\partial p}{\partial S}\right)_V (S_1 - S_0) = -\frac{1}{12T_0}\left(\frac{\partial p}{\partial S}\right)_V \left(\frac{\partial^2 p}{\partial V^2}\right)_S (V_1 - V_0)^3 \tag{171}$$

and since the terms $\left(\frac{\partial p}{\partial S}\right)_V > 0$ and $\left(\frac{\partial^2 p}{\partial V^2}\right)_S > 0$ for normal gases, the third derivative of the Hugoniot curve is less than the third derivative of the Isentrope curve at point A - which means that the Hugoniot curve passes below the Isentrope curve for $V_1 > V_0$ as shown in fig. 1.34.

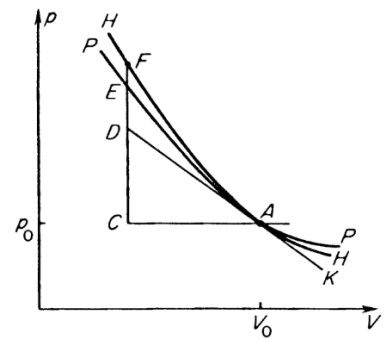


Fig. 1.34. Relative position of the shock *H* and ordinary isentrope *P*. *DK* is the tangent to the curves at the initial state *A*. In a weak shock wave the segment *CD* is a quantity of first, *DE* of second, and *EF* of third order.

This proof should also be validated in my understanding.

18.3 Geometric interpretation of the arguments

1. The book discusses (pg. 66 bottom - 67) the geometric interpretation of the arguments above using fig. 1.35 in the book.

The book says that the correction for the area of the triangle ABC to the entire area ABCEA which is equal $\bar{T}\Delta S$ (where $\bar{T}$ is some average temperature between T_0 and T_1) is of a higher order (i.e. ΔV^4) and therefore negligible for weak shocks. It suggests that one can see geometrically through expressing the area of ABCEA as:

$$\text{Area} = \bar{T}\Delta S = \frac{p_0 + p_1}{2}(V_0 - V_1) - \int_{V_1}^{V_0} P(V, S_0)dV \tag{172}$$

which is exactly the area of the trapezoid MABN (M,N don't appear in this figure so imagine them on the vertical lines from A and B to the V axis) minus the area under the isentrope curve from V_1 to V_0.

Then, the book says to substitute the third-order expansion for p along the isentrope curve to get the same relation (1.88) in the book. I asked chatGPT to do so, it might be a wonderful exercise to practice the Taylor expansions...

2. The book also compares the slopes of the secant line AB and the tangent line at points A,B. It does so by expanding these slopes to the first order in ΔV:

$$\frac{p_1 - p_0}{V_1 - V_0} = \left(\frac{\partial p}{\partial V}\right)_S + \frac{1}{2}\left(\frac{\partial^2 p}{\partial V^2}\right)_S (V_1 - V_0) + O(\Delta V^2) \tag{173}$$

$$\left(\frac{\partial p}{\partial V}\right)_{S_A} = \left(\frac{\partial p}{\partial V}\right)_{S_0} \tag{174}$$

$$\left(\frac{\partial p}{\partial V}\right)_{S_B} = \left(\frac{\partial p}{\partial V}\right)_{S_0} + \left(\frac{\partial^2 p}{\partial V^2}\right)_S (V_1 - V_0) + O(\Delta V^2) \tag{175}$$

where the tangent slope at point B was expanded using the expansion of the derivative $\left(\frac{\partial p}{\partial V}\right)_S$ along the isentrope curve around point A (where $S = S_0$). I need to understand this point better.

A very important quote of the book regarding to the deepness of the geometric interpretations:: "Of importance is the inner connection between the condition for entropy increase and that for mechanical stability of the discontinuity $u_0 > c_0$. (The geometric interpretation) Both conditions fol

18.4 Summary - The proven properties of weak shocks in normal fluids

The following are the actual properties that were proven about weak shocks in normal fluids (in which $\left(\frac{\partial^2 p}{\partial V^2}\right)_S > 0$):

(a) For a shock wave the entropy increases through the shock - $S_1 > S_0$ while for a rarefaction shock the entropy decreases through the shock - $S_1 < S_0$ (pg. 63-64 analytically and pg. 65 bottom - 66 topgeometrically). The entropy increase is proportional to the cube of the volume difference across the shock - $\Delta S \sim (V_1 - V_0)^3$ (e.g. very small for weak shocks).

(b) The Hugoniot and Isentrope curves are tangent up to second order at point A (pg. 64 bottom -65 top).

(c)

(d) For shock wave $u_0 > c_0$ (supersonic flow into the shock) & $u_1 < c_1$ (subsonic flow out of the shock) - and the other way around for rarefaction shocks (pg. 66 bottom - pg. 67 top geometrically).

19 Shock waves in a fluid with anomalous thermodynamic properties

This section is divided into two subsections:

1. Analyzing gases for which the condition for normal fluids is violated - e.g. their isentropes and Hugoniot curves are convex upwards in the $p - V$ plane. For these gases, rarefaction shocks are possible while compression shocks are impossible (for the reasons of entropy increase and mechanical stability explained before).

2. An example of such fluids are liquids near the critical point, where the coefficient of thermal expansion at constant pressure is negative - this is shown in fig. 1.37 and explained in the bottom of pg. 68 and slightly before the section ends.

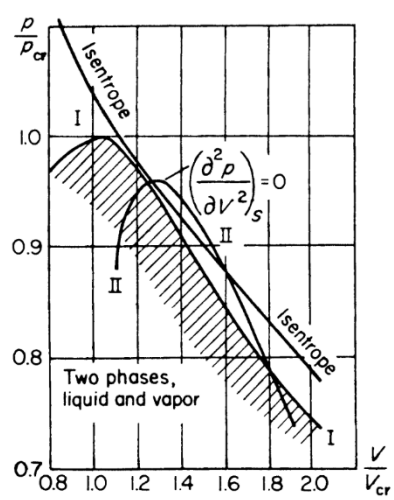


Fig. 1.37. Isentrope with anomalous convexity in a van der Waals' gas with a specific heat $c_v = 40$ cal/deg-mole. The shaded area is a two-phase region. Curve II bounds the region of anomalous convexity of the isentrope. Under curve II $(\partial^2 p/\partial V^2)_S < 0$.

20 Equations of 1D gas flow

20.1 Introduction

1. The book introduces the fact that dissipative processes are viscosity and heat conduction, and are both neglected in the Euler equations. These processes lead to **nonadiabatic and thermodynamically irreversible** phenomena in the gas flow, and they appear only when there are large gradients in the flow variables (like in shock waves).

2. The book says that we are going to deal with only the 1D planar solution.

20.2 Considering viscosity - derivation of the generalized momentun tensor

1. The book presents a cross section of the flow surrounded by two adjacent layers whose thickness is of the order of the mean free path l (fig. 1.38 in the book) - in the volumes of these layers, the particles are asssumed to have a constant velocity u (since we suppose that a change in velocity happens through collision, which happens for each molecule only after traveling a distance of the order of l).

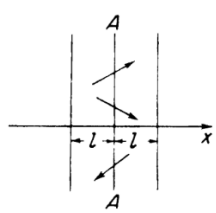


Fig. 1.38. Schematic illustration for the derivation of the equations for the molecular transport of momentum.

2. Then, the book analyzes the amount of pressure that is exerted on the cross section by the particles in the two layers. It turns out that this pressure equals to the rate of momentum transfer per unit area from the two layers to the cross section. For that we define m as the mass of each particle, n as the number density of particles, and ν as the average velocity of particles in each layer (which is of the order of the thermal velocity). The rate of momentum transfer from the left layer to the cross section is:

$$\text{Rate of momentum transfer from left layer} = n\nu \cdot mu \tag{176}$$

while the rate of momentum transfer from the right layer to the cross section is:

$$\text{Rate of momentum transfer from right layer} = -n\nu \cdot m(u+\Delta u) = -n\nu \cdot mu - n\nu \cdot ml\frac{\partial u}{\partial x} \tag{177}$$

Therefore, the total rate of momentum transfer to the cross section per unit area and unit time is:

$$\text{Total rate of momentum transfer} = -n\nu ml\frac{\partial u}{\partial x} \tag{178}$$

Therefore, the viscous stress (the extra pressure due to viscosity) is:

$$\tau_{xx} = -\sigma' = -\frac{4}{3}\mu\frac{\partial u}{\partial x} \tag{179}$$

where $\frac{4}{3}$ is a numerical factor that comes from a more accurate kinetic theory calculation for an ideal gas, and $\mu \propto nm\nu l = \rho\nu l$ is the dynamic viscosity coefficient of the gas. The chatGPT says that viscosity is rate of momentum diffusion that arises from the velocity gradients in the fluid, which physically means that particles from faster layers transfer momentum to slower layers through collisions, which is exactly what we analyzed here. In contrast, thermodynamic pressure arises from the random isotropic motion of particle that exists even in the a unifrom flow without velocity gradients (e.g. equillibrium).

3. Therefore, the generalized momentum flux tensor that includes the effect of viscosity is:

$$\Pi_{xx} = p - \sigma' = p - \frac{4}{3}\mu\frac{\partial u}{\partial x} \tag{180}$$

and the continuity equation is:

$$\frac{\partial}{\partial t}(\rho u) = -\frac{\partial \Pi_{xx}}{\partial x} \tag{181}$$

$$\rho\frac{Du}{Dt} = -\frac{\partial}{\partial x}(p - \sigma') \tag{182}$$

and the book says that $\frac{\partial \sigma}{\partial x}$ is the viscous (e.g. internal friction) force per unit volume that acts to reduce velocity gradients in the fluid.

20.3 Considering Heat conduction - generalized energy flux

1. The effective decrement in momentum flux due to viscosity leads to a decrease in the kinetic energy of the fluid, which is converted to internal energy (heating) through viscous dissipation. Therefore, a term of $\sigma' u$ analogous to pu should be added to the energy flux.

2. The energy flux per unit area and unit time due to heat conduction is given by Fourier's law:
$$J = -\kappa \frac{\partial T}{\partial x} \tag{183}$$
where κ is the thermal conductivity of the gas and is of the order of $\rho c_p \nu l$.

3. Therefore, the generalized energy flux that includes the effects of viscosity and heat conduction is:
$$\frac{\partial}{\partial t}\left[\rho\left(\varepsilon + \frac{u^2}{2}\right)\right] = -\frac{\partial}{\partial x}\left[\rho u\left(\varepsilon + \frac{u^2}{2}\right) + (p - \sigma')u - \kappa\frac{\partial T}{\partial x}\right] \tag{184}$$
which can be transformed with the aid of the continuity & momentum equations (and also the second law of thermodynamics) to the following equation:
$$\rho T \frac{DS}{Dt} = \sigma' \frac{\partial u}{\partial x} - \frac{\partial J}{\partial x} = \frac{4}{3}\mu\left(\frac{\partial u}{\partial x}\right)^2 + \frac{\partial}{\partial x}\left(\kappa\frac{\partial T}{\partial x}\right) \tag{185}$$
The book comments that the viscosity term is always positive and represents the irreversible conversion of kinetic energy to internal energy (heating) due to viscous dissipation, while the heat conduction term can be positive or negative depending on the temperature gradient.

4. Dispite the last comment, the book emphasizes that both viscosity and heat conduction lead to an increase in entropy in the fluid (e.g. irreversible processes) as should be expected from the second law of thermodynamics. This can be seen through the integration of the heat conduction term over a volume bounded by two surfaces at $x = x_1 < x_2$ - which leads to a positive contribution to the total entropy change in the volume.

20.4 Definition of Reynolds & Peclet numbers

1. These two dimensionless numbers are defined in order to compare the relative importance of the dissipative processes (viscosity and heat conduction) to the advective transport of momentum and energy in the fluid.

2. If we suppose that U is the characteristic velocity & d is the characteristic length scale of the flow region - we can claim that:

$$\varepsilon_{\text{advection}} = \rho u \frac{\partial u}{\partial x} \sim \frac{\rho U^2}{d} \tag{186}$$

$$\varepsilon_{\text{viscosity}} = \frac{\partial \sigma'}{\partial x} = \frac{4}{3}\mu \frac{\partial^2 u}{\partial x^2} \sim \frac{\mu U}{d^2} \tag{187}$$

$$\varepsilon_{\text{heat conduction}} = \frac{\partial J}{\partial x} = \frac{\partial}{\partial x}\left(-\kappa \frac{\partial T}{\partial x}\right) \sim \frac{\kappa \Delta T}{d^2} \sim \frac{\kappa U}{c_p d^2} \tag{188}$$

The Reynolds and Peclet numbers are defined as the ratios of advective transport to dissipative transport:

$$Re = \frac{\varepsilon_{\text{advection}}}{\varepsilon_{\text{viscosity}}} = \frac{\rho U^2/d}{\mu U/d^2} = \frac{\rho U d}{\mu} = \frac{\nu}{U d} \tag{189}$$

$$Pe = \frac{\varepsilon_{\text{advection}}}{\varepsilon_{\text{heat conduction}}} = \frac{\rho U^2/d}{\kappa U/c_p d^2} = \frac{\rho c_p U d}{\kappa} = \frac{\chi}{U d} \tag{190}$$

where $\nu = \frac{\mu}{\rho}$ is the kinematic viscosity and $\chi = \frac{\kappa}{\rho c_p}$ is the thermal diffusivity of the gas.

3. The book comments that $\chi, \nu \sim \nu l$ by definition (they differ by a proportionality constant of order unity). Therefore, the Reynolds and Peclet numbers are of the same order of magnitude.

4. In flows where $Re, Pe \gg 1$, the dissipative processes are negligible compared to the advective transport of momentum and energy. However, in regions with large gradients in the flow variables (like shock waves), the dissipative processes become significant even if $Re, Pe \gg 1$ globally because the characteristic length scale $fracdl$ becomes very small in these regions.